

Can Machine Learning Outperform Professional Consensus Forecasts in Forecasting EM Sovereign Yields?

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Outline

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Introduction

What Is a Sovereign Yield?

The interest rate a government pays to borrow money

Think of it as the “price of trust”

Higher yield $\Rightarrow$ market demands more compensation $\Rightarrow$ **higher perceived risk**

Three maturities in this paper:

3M	Money market	Short-term liquidity
2Y	Policy path	Central bank expectations
10Y	Risk premium	Long-run global risk sentiment

The challenge

EM sovereign yields are driven by **both** domestic fundamentals **and** global financial shocks — hard to forecast.

Motivation

Professional consensus forecasts are the market standard

Aggregated from major banks & institutions — used by portfolio managers, policy analysts, and central banks.

But are they accurate?

- If consensus forecasts are **systematically biased**...
- ...can ML **identify and correct** these biases out of sample?

Our approach: Forecast Error Correction

Instead of predicting yield *levels*, we predict and correct the **errors** in consensus forecasts.

Evidence of Systematic Bias

Coibion & Gorodnichenko (2015) diagnostic:

Under rational expectations, forecast revisions should *not* predict errors:

$$e_{i,m,t+h} = \alpha + \beta \cdot r_{i,m,t} + \varepsilon$$

Horizon	$\hat{\beta}$	Significance
3M	-0.894	***
6M	-0.318	
9M	-0.339	
12M	-0.497	*
18M	-0.636	**
24M	-0.573	**

Forecasters overshoot when revising, rational expectations **rejected**.

Systematic overreaction $\Rightarrow$ **learnable patterns** $\Rightarrow$ ML can help!!!

Literature

Related Literature

Forecast Error & Belief Distortions

- Coibion & Gorodnichenko (2015)
Information rigidity in professional forecasts
- Bianchi, Ludvigson & Ma (2022)
Belief distortions as systematic biases in macro forecasting

Emerging Market Sovereign Yields

- Jaramillo & Weber (2013)
U.S. rate shocks $\rightarrow$ state-dependent EM yield transmission
- Csontó & Ivaschenko (2013)
VIX dominates short-run EM sovereign spreads
- IMF (2025)
 $VIX \uparrow 10pp \Rightarrow$ EM yield spreads widen significantly

Related Literature

ML in Macroeconomic Forecasting

- Coulombe et al. (2022)
Gradient boosting outperforms sparsity-based models
- Makridakis et al. (2022)
LightGBM dominates M5 forecasting competition
- Nunes et al. (2026)
LSTM-LagLasso for sovereign yield forecasting

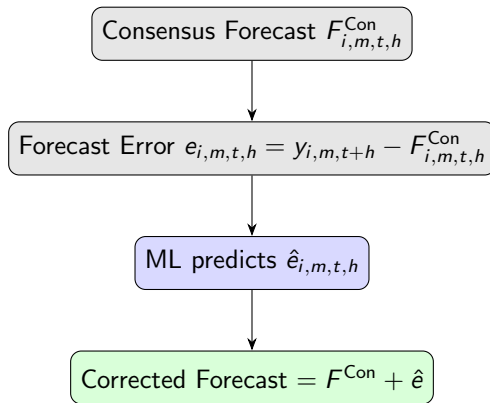
This paper's contribution

- First systematic ML vs. consensus comparison for EM sovereign yields
- “Three mechanisms, three models” framework
- VIX-triggered model selection rule for practitioners

Data & Methodology

What We Do

Forecast Error Correction Framework:



Three estimated models: ARX, LightGBM, LSTM

Dataset: 16 EM economies $\times$ 3 tenors $\times$ 6 horizons, 2009 Q1–2025 Q2,
 $N = 3,662$

Data — Panel Structure

16 Emerging Market Economies:

Bulgaria	Czech Rep.	Hungary	Poland
Romania	Russia	Turkey	Brazil
Mexico	India	Indonesia	Malaysia
Philippines	South Africa	Thailand	Taiwan

Sample: 2009 Q1 – 2025 Q2 ($N = 3,662$ usable observations)

Three maturities: 3M, 2Y, 10Y

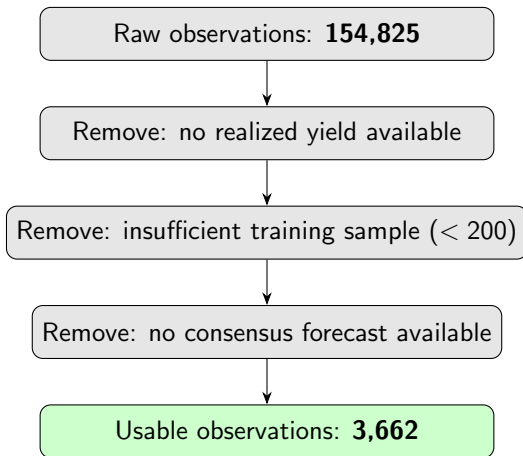
Six forecast horizons: 3M, 6M, 9M, 12M, 18M, 24M

Unbalanced panel

Not all country $\times$ tenor $\times$ horizon combinations are observed — handled via equal weighting (primary) and inverse frequency weighting (robustness).

Data Construction (Nobody cares but it took me 9 months)

From raw observations to usable panel:



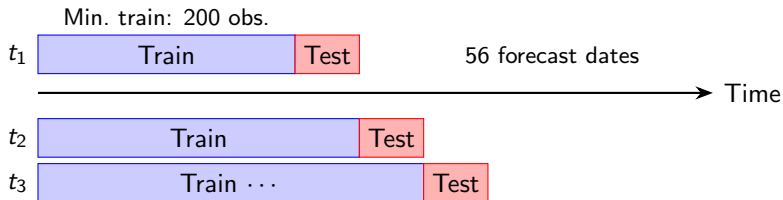
Feature Set

Category	Variables
Consensus forecasts	bbg_consensus_tph, bbg_consensus_lag1, forecast_revision
Current yield	current_yield_t
U.S. rates	us_10y_rate, us_2y_rate
Domestic macro	GDP, CPI, CA forecasts (local)
U.S. macro	GDP, CPI, CA forecasts (U.S.)
Global financial	VIX_Index, SPX_Index, C01-Comdty, fx-usd-local
Structure dummies	Tenor dummies, horizon dummies

Missing values: median imputation (train set) + binary missing indicators.

Methodology — Rolling Window Design

Expanding window out-of-sample evaluation:



- No look-ahead bias: **test set never seen during training**
- Median imputation computed on train set only

Three Estimated Models (1/2)

Model 1: ARX (Linear Benchmark)

- OLS regression with full feature set
- Core predictor: `forecast_revision` — captures linear overreaction
- Re-estimated at each rolling window; no hyperparameters

Model 2: LightGBM (Non-linear)

- Gradient-boosted decision trees
- Captures interaction effects & threshold non-linearities
- Hyperparameters selected via time-series CV at each rolling window
 - Search space: $\text{learning rate} \in \{0.01, 0.05, 0.10\}$, $\text{num_leaves} \in \{15, 31, 63\}$, $\text{n_estimators} \in \{100, 200, 500\}$
 - 27 combinations; validation = last 20% of training set

Three Estimated Models (2/2)

Model 3: LSTM (Deep Learning)

- Long short-term memory network
- Input: sequence of last $L = 4$ quarters per observation
- Output: predicted forecast error $\hat{e}_{i,m,t,h}$
- Fixed hyperparameters (hidden = 64, layers = 2, dropout = 0.2)
- Early stopping (patience = 10)

Robustness: Ablation Study on Sequence Length

Sequence length $L \in \{2, 4, 6\}$ tested — $L = 4$ adopted as primary specification.

Results reported in results section.

Evaluation Metric

Out-of-Sample R^2 (OOS R^2):

$$\text{OOS } R^2 = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{consensus}}}$$

- OOS $R^2 > 0 \Rightarrow$ model **beats** the consensus
- OOS $R^2 = 0 \Rightarrow$ model **ties** the consensus
- OOS $R^2 < 0 \Rightarrow$ model **worse** than consensus

Statistical significance tests:

Test	Use case	Why
Clark-West (2007)	Model vs. consensus	Nested comparison
Diebold-Mariano (1995)	Model vs. model	Non-nested comparison

Block bootstrap ($b = 8$, $B = 1,000$) for 95% confidence intervals.

Results

Main Results — Overall

	Consensus	ARX	LightGBM	LSTM
OOS R^2	0.000	0.374	0.490	0.470
RMSE (bps)	209.6	165.9	149.7	139.0
MAE (bps)	108.5	115.4	83.0	73.1

All three models outperform the consensus!!!

- LightGBM saves **~60 bps** per forecast vs. consensus on RMSE
- ARX saves **~44 bps** per forecast vs. consensus on RMSE

Main Results — By Maturity

Tenor	Consensus	ARX	LightGBM	LSTM
10Y	0.000	0.245	0.480	0.349
2Y	0.000	0.568	0.496	0.794
3M	0.000	0.485	0.519	0.733

No single model dominates across all maturities:

- **10Y** — LightGBM best: static non-linear interactions dominate
- **2Y** — LSTM best: cross-period temporal dynamics dominate
- **3M** — LSTM best: rich temporal structure at long horizons

Key finding

Different maturities are governed by **different structures** $\Rightarrow$ different models work best for different maturities.

Main Results — By Forecast Horizon

Horizon	Consensus	ARX	LightGBM	LSTM
3M	0.000	0.557	0.341	0.191
6M	0.000	0.272	0.449	0.321
9M	0.000	0.342	0.529	0.481
12M	0.000	0.325	0.525	0.496
18M	0.000	0.363	0.620	0.706
24M	0.000	0.280	0.529	0.528

- **Short (3M)**: linear overreaction dominates → ARX strongest
- **Medium (6M–12M)**: non-linear structure emerges → LightGBM strongest
- **Long (18M)**: temporal persistence dominates → LSTM strongest

Tenor $\times$ Horizon OOS R^2 — ARX

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	0.376	0.098	0.182	0.210	0.326	0.222
2Y	0.688	0.536	0.557	0.504	0.246	0.172
3M	0.287	0.274	0.530	0.502	0.535	0.493

- **All 18 cells positive** — ARX beats consensus everywhere
- Weakest: 10Y $\times$ 6M (0.098)
- Strongest: 2Y $\times$ 3M (0.688)

Tenor $\times$ Horizon OOS R^2 — LightGBM

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	0.358	0.408	0.467	0.498	0.577	0.511
2Y	0.358	0.603	0.669	0.538	0.710	0.604
3M	-0.224	0.058	0.457	0.621	0.703	0.556

- **17/18 cells positive**
- Only negative: 3M tenor $\times$ 3M horizon (**-0.224**) — probably with high noise, model overfits
- Strongest: 2Y $\times$ 18M (0.710)

Tenor $\times$ Horizon OOS R^2 — LSTM

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	-0.316	0.310	0.452	0.450	0.661	0.435
2Y	0.919	0.202	0.480	0.526	0.789	0.752
3M	0.123	0.580	0.705	0.715	0.803	0.732

- **16/18 cells positive**
- Negative: 10Y $\times$ 3M (**-0.316**) — no temporal persistence in short-horizon 10Y errors
- **Strongest: 2Y $\times$ 3M (0.919)** — cross-period memory most powerful here

Statistical Significance — Clark-West Test

Nested model comparison: each model vs. consensus benchmark

$H_0 : \text{OOS } R^2 = 0$ (model performs no better than consensus)

Model	CW Statistic	Significance
ARX	9.32	***
LightGBM	10.82	***
LSTM	8.60	***

Interpretation

Positive CW statistic $\Rightarrow$ model beats the consensus benchmark.
All three models reject H_0 at the 1% level across almost all maturity and horizon subsamples.

Statistical Significance — Diebold-Mariano Test

Non-nested model comparison: pairwise

$$d_t = e_{\text{model 1}}^2 - e_{\text{model 2}}^2 \quad \text{DM} = \frac{\bar{d}}{\text{se}(\bar{d})}$$

Comparison	DM Statistic	Significance
LightGBM vs. ARX	−6.33	***
LSTM vs. ARX	−2.39	**
LSTM vs. LightGBM	+0.18	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Two-sided test.

Interpretation

Negative DM $\Rightarrow$ first-named model has **lower MSE** (performs better)

Positive DM $\Rightarrow$ second-named model has lower MSE

Not significant $\Rightarrow$ two models statistically indistinguishable

Statistical Significance — Key Takeaway

Finding 1: All models beat the consensus

Clark-West test confirms all three models significantly outperform the consensus benchmark ($p < 0.01$) — ruling out chance.

Finding 2: LightGBM significantly beats ARX

DM statistic = -6.33 ($p < 0.01$) — non-linear structure delivers statistically significant incremental gains over the linear model.

Finding 3: LSTM $\approx$ LightGBM overall, but not everywhere

Statistically indistinguishable overall ($p = 0.86$)
— but LSTM **significantly outperforms** LightGBM in the **3M maturity segment** ($p < 0.01$),
where temporal memory adds unique value.

Consensus-Level Benchmark (CLB)

Does LightGBM just exploit a simple linear relationship between consensus level and forecast errors?

We test: a linear model using *only* bbg_consensus_tph:

$$\hat{e}_{i,m,t,h} = \alpha + \beta \cdot F_{i,m,t,h}^{\text{Con}}$$

Model	Full Sample ($N = 3,437$)	Common Sample ($N = 2,973$)
CLB (consensus level only)	0.023	-0.070
ARX	0.374	0.264
LightGBM	0.490	0.480
LSTM	—	0.468

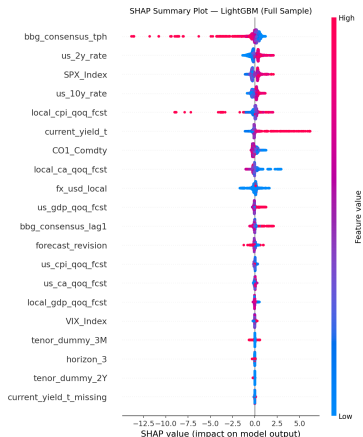
Answer: No

CLB is near zero or **negative**.

Gains come from the **broader feature set** and **non-linear interactions**, not the consensus level alone.

SHAP & Crisis

SHAP Analysis — Full Sample



- **bbg_consensus_tph**: top feature — non-linear overreaction signal
- **us_2y_rate**: U.S. rate expectations → EM yield underprediction
- **SPX_Index**: global risk sentiment — heterogeneous across maturities

SHAP Analysis — By Maturity

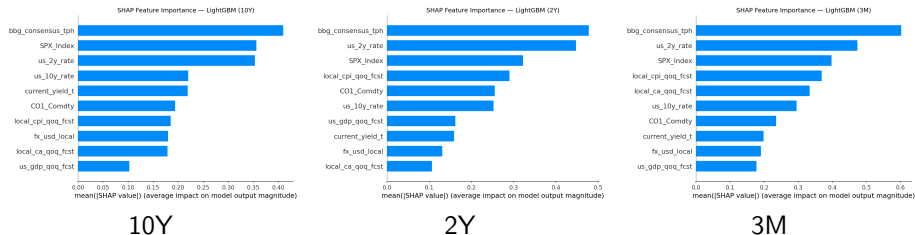
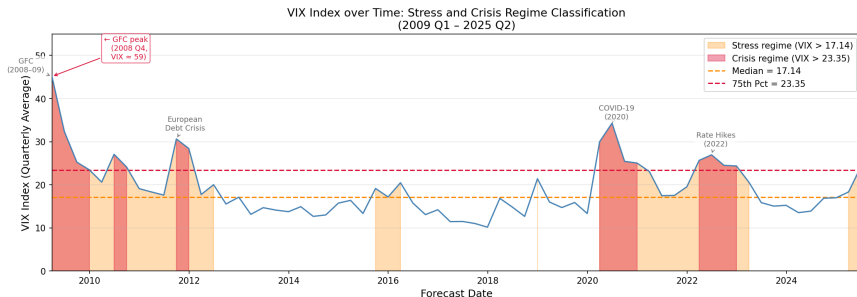


Figure: LightGBM SHAP Feature Importance by Tenor (Top 10). Bars show mean absolute SHAP values.

- **10Y:** SPX ranks 2nd — long-tenor yields most sensitive to risk sentiment
- **2Y:** local CPI forecast enters top 5 — policy reaction function channel
- **3M:** most evenly distributed — no single factor dominates

Crisis vs. Non-Crisis — VIX Regime Classification



- **Stress regime:** $VIX > 17.14$ (sample median)
- **Crisis regime:** $VIX > 23.35$ (75th percentile)

Crisis vs. Non-Crisis — Results

Model	Stress regime		Crisis regime	
	Low VIX	High VIX	Normal	Crisis
ARX	0.432	0.311	0.445	0.048
LightGBM	0.435	0.548	0.517	0.367
LSTM	0.363	0.532	0.545	0.208

- **Low stress:** ARX sufficient — linear patterns dominate
- **Moderate stress:** ML gains largest — non-linearities most active
- **Extreme crisis:** all models deteriorate — consensus itself unstable

Sweet spot

ML delivers the most value when VIX is **moderately elevated** ($17 < \text{VIX} \leq 23$) — not in extreme crises.

Discussion

Three Mechanisms, Three Models

Maturity	Pricing Mechanism	Dominant Structure	Best Model
3M	Money market	Linear overreaction	ARX
2Y	Policy path	Cross-period persistence	LSTM
10Y	Risk premium	Static non-linearities	LightGBM

Why no single model dominates

Different maturities are governed by **fundamentally different structures** — each requiring a different model architecture to exploit.

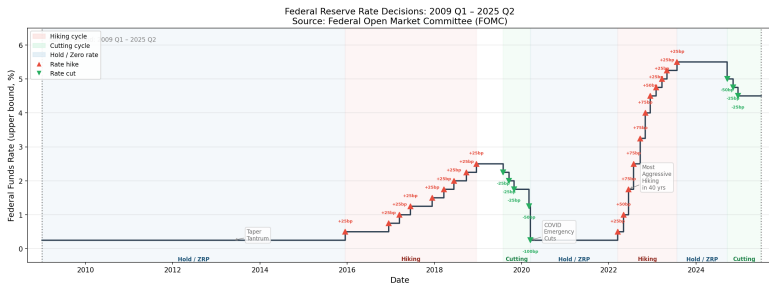
From empirical finding to economic interpretation

3M: forecast revision already captures linear overreaction

2Y: policy cycles persist across quarters → memory mechanism most valuable

10Y: $VIX \times SPX \times$ consensus interactions → gradient boosting most valuable

VIX-Triggered Model Selection Rule



VIX Level	Regime	Recommendation
≤ 17.14	Low stress	ARX sufficient; trust consensus
$17.14 < \text{VIX} \leq 23.35$	Moderate stress	Activate LightGBM or LSTM
> 23.35	Extreme stress	Use all models conservatively

Limitations & Future Research

Limitations:

- 16 EM economies — Latin America and Africa underrepresented
- Sample starts post-GFC; excludes 2008 crisis core phase
- Russia 2022+ data reliability concerns
- LSTM uses fixed hyperparameters (vs. LightGBM CV tuning)
- No textual information (Fed FOMC statements, meeting minutes)

Future Research:

- Broader geographic coverage (Africa, Middle East)
- NLP features from Fed communications
- Transformer & hybrid LightGBM-LSTM architectures
- Dynamic model ensembles triggered by VIX
- Apply framework to equities & exchange rates

Conclusion

Conclusion

❶ **Consensus forecast errors are systematically predictable.**

CG (2015) diagnostic confirms overreaction ($\hat{\beta} < 0$) at short and long horizons — rational expectations rejected.

❷ **All three ML models significantly outperform the consensus.**

OOS R^2 : ARX 0.374, LightGBM 0.490, LSTM 0.470
(Clark-West test, $p < 0.01$; saving 44–60 bps per forecast)

❸ **Non-linear structure and temporal dynamics add incremental value.**

LightGBM nearly doubles ARX's OOS R^2 in 10Y segment.
LSTM achieves OOS $R^2 = 0.794$ in 2Y segment.

❹ **No single model dominates — optimal choice varies by maturity.**

Three mechanisms, three models framework.
VIX-triggered switching rule for practitioners.

Thank You

Questions & Discussion

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Data provided by Chi Chun-Che & Kenji Wada